

Tableau Assignment – Sample Superstore Dataset

Q1. What is Tableau and why is it preferred over traditional spreadsheet tools for data analysis?

Answer:

Tableau is a powerful data visualization and business intelligence tool used to analyze, visualize, and present data in an interactive format. It helps users convert raw data into meaningful insights using charts, graphs, and dashboards.

Tableau is preferred over traditional spreadsheet tools like Excel because:

- It can handle large datasets efficiently without slowing down.
- It provides interactive dashboards instead of static tables.
- It allows drag-and-drop analysis without complex formulas.
- It connects to multiple data sources easily.
- It updates visualizations dynamically when data changes.

Industries using Tableau:

1. **Retail Industry**
2. **Healthcare Industry**

Use Case (Retail Industry):

A retail company uses Tableau to analyze sales performance across regions. It helps identify best-selling products, customer buying patterns, and underperforming areas, enabling better marketing and inventory decisions.

Q2. Sample – Superstore Dataset

a) Main Types of Fields

After connecting to the Sample Superstore dataset, the following types of fields are available:

- Time-based fields (Order Date, Ship Date)
 - Geographical fields (State, City, Region, Country)
 - Numerical fields (Sales, Profit, Quantity, Discount)
 - Categorical fields (Category, Sub-Category, Segment, Customer Name)
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b) Dimensions and Measures

Dimensions (Blue Fields):

1. Category

2. Sub-Category
3. Region
4. Customer Name

Measures (Green Fields):

1. Sales
 2. Profit
 3. Quantity
 4. Discount
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Q3. Create a Bar Chart Showing Total Sales by Category

Answer:

To create a bar chart in Tableau:

- Drag **Category** to the **Columns** shelf.
- Drag **Sales** to the **Rows** shelf.

Explanation:

- Columns shelf: Category (Dimension)
- Rows shelf: Sales (Measure)
- Aggregation applied: **SUM(Sales)** (default aggregation in Tableau)

Conclusion:

This chart shows total sales for each category, helping compare performance across categories.

Q4. Insight from Sales and Profit & Profit by Sub-Category Chart

Answer:

Comparing Sales and Profit helps identify whether high sales are actually generating profit. A product may have high sales but low profit due to discounts or high costs.

Steps to create Profit by Sub-Category chart:

- Drag **Sub-Category** to Rows.
- Drag **Profit** to Columns.
- Tableau automatically applies **SUM(Profit)**.

Insight:

Some sub-categories generate high profit, while others may cause losses.

Conclusion:

This analysis helps identify profitable and unprofitable product segments.

Q5. Line Chart – Sales Trend over Order Date**Answer:**

To create a line chart:

- Drag **Order Date** to Columns.
- Drag **Sales** to Rows.
- Change date level to **Month**.

Best Date Level: Month

Reason:

Month level shows detailed trends and seasonal patterns better than Year or Quarter.

Why Line Chart is suitable:

- Shows trends over time clearly
 - Helps identify growth or decline
 - Useful for seasonal analysis
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Q6. Apply Basic Filter (Region = West, Category = Technology)**Answer:****Steps:**

- Drag **Region** to the Filters shelf and select **West**
- Drag **Category** to the Filters shelf and select **Technology**

Shelf Used:

- Filters shelf

Conclusion:

The view will only show data for West region and Technology category.

Q7. Importance of Geographical Analysis & Map Creation**Answer:**

Geographical analysis helps businesses understand performance across different locations. It helps identify strong and weak regions, optimize distribution, and improve decision-making.

Steps to create Map:

- Double-click **State**
- Drag **Sales** to Color or Size

Conclusion:

Darker areas show higher sales, helping identify high-performing states.

Q8. Worksheet vs Dashboard in Tableau

Answer:

A worksheet contains a single visualization, while a dashboard combines multiple worksheets into one view.

Why dashboard is better:

- Shows multiple insights in one screen
 - Helps in faster decision-making
 - Allows interactive filtering
 - Provides a complete business overview
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Q9. What is Tableau Prep and its importance?

Answer:

Tableau Prep is a data cleaning tool used to prepare raw data for analysis.

Importance:

- Improves data quality
- Removes errors and inconsistencies
- Saves time in data cleaning
- Makes analysis more accurate

Steps:

1. Remove Null Values:

- Filter out NULL rows using Clean Step

2. Remove Duplicates:

- Use Remove Duplicates option in Clean Step

3. Fix Data Types:

- Change field type (e.g., text to number/date)
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Q10. Retail Manager Business Questions

a) Charts Used:

- Highest profit category → Bar Chart
 - Poor-performing region → Map or Bar Chart
 - Sales over time → Line Chart
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b) Key Insight:

Technology may generate highest profit, while some regions may underperform.

c) Business Decision Impact:

- Helps identify profitable and weak areas
- Improves marketing strategies
- Supports better inventory planning
- Enables data-driven decision-making